

LIPID PRACTICE QUESTIONS

1. Which characteristic primarily defines the heterogeneous group of molecules known as lipids?

- A. Presence of nitrogenous bases
- B. High solubility in water
- C. Hydrophobicity (water-insolubility)
- D. Linear polymer structure of amino acids

Answer: C

2. Lipids can be extracted from tissues using which type of solvents?

- A. Polar solvents
- B. Nonpolar solvents
- C. Aqueous solutions
- D. Buffered solutions

Answer: B

3. Which of the following is a primary function of body lipids?

- A. Providing a hydrophobic barrier for partitioning cellular contents
- B. Serving as the primary repository of genetic information
- C. Acting as the sole source of blood glucose during a fast
- D. Catalyzing all biochemical reactions as enzymes

Answer: A

4. How are lipids typically transported in the aqueous environment of the plasma?

- A. As free, dissolved molecules
- B. Associated with carbohydrates in micelles
- C. In association with proteins, such as albumin or lipoproteins
- D. Bound to hemoglobin

Answer: C

5. Which type of lipid serves as a major energy reserve for the body?

- A. Phospholipids
- B. Cholesterol
- C. Triacylglycerols in white adipocytes

D. Glycolipids

Answer: C

6. Which molecules are the most reduced and provide the highest energy yield upon oxidation (9 kcal/g)?

A. Carbohydrates

B. Proteins

C. Triacylglycerols

D. Nucleic acids

Answer: C

7. Body lipids are generally found "compartmentalized." An example of this is:

A. Free glucose in the blood

B. Membrane-associated lipids

C. Dissolved amino acids in the cytosol

D. Soluble vitamins in the plasma

Answer: B

8. The "amphipathic" nature of certain lipids, like fatty acids, refers to:

A. Having only hydrophobic regions

B. Having both a hydrophilic and a hydrophobic region

C. Being soluble in both water and oil

D. Having four fused hydrocarbon rings

Answer: B

9. Which of the following is NOT considered a major class of lipids in the human body?

A. Fatty acids

B. Phosphoacylglycerols

C. Polysaccharides

D. Sphingolipids

Answer: C

10. The hydrophobic effect that causes lipids to cluster in water is driven by:

A. Covalent bonding between lipid tails

B. Water molecules maximizing hydrogen bonds with each other

C. Ionic attraction between head groups

D. The presence of enzymes in the water

Answer: B

11. Simple lipids are primarily defined as esters of fatty acids with:

- A. Various alcohols
- B. Amino acids
- C. Phosphate groups
- D. Carbohydrates

Answer: A

12. Triacylglycerols (triglycerides) are classified as simple lipids consisting of three fatty acids esterified to:

- A. Sphingosine
- B. Glycerol
- C. Cholesterol
- D. Inositol

Answer: B

13. Which of the following is a "compound" or complex lipid?

- A. Palmitic acid
- B. Cholesterol
- C. Phosphatidylcholine
- D. Acetoacetate

Answer: C

14. Complex lipids like phospholipids contain an alcohol attached by a phosphodiester bridge to:

- A. Either diacylglycerol or sphingosine
- B. A steroid nucleus
- C. A single fatty acid
- D. A protein core

Answer: A

15. Glycolipids (glycosphingolipids) are compound lipids that contain:

- A. Phosphate groups
- B. Carbohydrates attached to a ceramide

- C. Three glycerol backbones
- D. High amounts of Vitamin D

Answer: B

16. Derived lipids are substances produced from the hydrolysis of simple and complex lipids. An example is:

- A. Chylomicrons
- B. Fatty acids
- C. VLDL
- D. Phosphatidic acid

Answer: B

17. Cholesterol is a derived lipid characterized by which structure?

- A. A linear aliphatic chain
- B. A four-fused-ring steroid nucleus
- C. A glycerol backbone
- D. A repeating disaccharide unit

Answer: B

18. Which of the following is a derived lipid that acts as a hormone-like signaling molecule?

- A. Triacylglycerol
- B. Eicosanoids
- C. Sphingomyelin
- D. Starch

Answer: B

19. Bile salts, which are derived from cholesterol, function to:

- A. Store energy in adipose tissue
- B. Emulsify dietary lipids in the intestine
- C. Transport oxygen in the blood
- D. Act as structural components of the cell wall

Answer: B

20. Ketone bodies, such as acetoacetate, are derived from the oxidation of:

- A. Glucose
- B. Amino acids

- C. Fatty acids
- D. Cholesterol

Answer: C

21. A fatty acid consists of a hydrophobic hydrocarbon chain and a terminal:

- A. Amino group
- B. Carboxyl group
- C. Phosphate group
- D. Hydroxyl group

Answer: B

22. What is the pKa of the terminal carboxyl group of a fatty acid?

- A. 2.0
- B. 4.8
- C. 7.4
- D. 9.5

Answer: B

23. At physiologic pH (~7.4), the terminal carboxyl group of a fatty acid is:

- A. Protonated ($-\text{COOH}$)
- B. Ionized ($-\text{COO}^-$)
- C. Attached to a glucose molecule
- D. Highly water-soluble

Answer: B

24. Most fatty acids in humans have an even number of carbon atoms, typically between:

- A. 2 and 6
- B. 4 and 10
- C. 16 and 20
- D. 30 and 40

Answer: C

25. Saturated fatty acids are characterized by:

- A. Having one or more double bonds
- B. Having only single bonds between carbons
- C. Being liquid at room temperature

D. Containing a phosphate group

Answer: B

26. Which of the following is a common 16-carbon saturated fatty acid?

A. Stearic acid

B. Palmitic acid

C. Oleic acid

D. Arachidonic acid

Answer: B

27. A fatty acid with one double bond is classified as:

A. Saturated

B. Monounsaturated

C. Polyunsaturated

D. A trans-fatty acid

Answer: B

28. Polyunsaturated fatty acids (PUFAs) contain:

A. No double bonds

B. Exactly one double bond

C. Two or more double bonds

D. Only triple bonds

Answer: C

29. In naturally occurring unsaturated fatty acids, double bonds are nearly always in the:

A. Trans configuration

B. Cis configuration

C. Saturated configuration

D. Alpha configuration

Answer: B

30. The "kink" in a fatty acid chain is caused by the presence of a:

A. Long carbon chain

B. Carboxyl group

C. Cis double bond

D. Trans double bond

Answer: C

31. What happens to the melting temperature (T_m) of a fatty acid as the chain length increases?

- A. It decreases
- B. It increases
- C. It stays the same
- D. It becomes zero

Answer: B

32. What happens to the melting temperature (T_m) of a fatty acid as the degree of unsaturation increases?

- A. It decreases
- B. It increases
- C. It stays the same
- D. It fluctuates randomly

Answer: A

33. In a polyunsaturated fatty acid, double bonds are typically spaced at:

- A. Two-carbon intervals
- B. Three-carbon intervals
- C. Six-carbon intervals
- D. The end of the chain only

Answer: B

34. Trans-fatty acids are primarily produced during:

- A. Natural synthesis in the liver
- B. Fasting in the adipose tissue
- C. Industrial hydrogenation of vegetable oils
- D. Digestion by pancreatic lipase

Answer: C

35. Which of the following is a 20-carbon fatty acid with four double bonds?

- A. Linoleic acid
- B. Oleic acid
- C. Arachidonic acid

D. Palmitic acid

Answer: C

36. In the standard numbering system for fatty acids, which carbon is Carbon 1?

- A. The methyl end carbon
- B. The carboxyl carbon
- C. The carbon next to the carboxyl group
- D. The middle carbon

Answer: B

37. The carbon atom to which the carboxyl group is attached (Carbon 2) is also called the:

- A. ω -carbon
- B. α -carbon
- C. β -carbon
- D. γ -carbon

Answer: B

38. The terminal methyl carbon of a fatty acid chain is always called the:

- A. α -carbon
- B. β -carbon
- C. ω -carbon (or n-carbon)
- D. Delta carbon

Answer: C

39. In the notation 18:1, Δ 9, what does the "1" represent?

- A. The position of the double bond
- B. The number of carbon atoms
- C. The number of double bonds
- D. The number of carboxyl groups

Answer: C

40. In the notation 20:4(5,8,11,14), the numbers in parentheses indicate:

- A. The total number of hydrogens
- B. The positions of double bonds relative to the carboxyl end
- C. The positions of methyl groups
- D. The melting temperature

Answer: B

41. The ω -nomenclature system counts carbons starting from the:

- A. Carboxyl end
- B. α -carbon
- C. Methyl-terminal end
- D. First double bond

Answer: C

42. If a fatty acid is described as an ω -3 fatty acid, this means:

- A. It has 3 double bonds
- B. It has 3 carbons
- C. The first double bond is 3 carbons from the methyl end
- D. It has 3 carboxyl groups

Answer: C

43. Oleic acid (18:1, Δ 9) is classified as which type of ω fatty acid?

- A. ω -3
- B. ω -6
- C. ω -9
- D. ω -7

Answer: C

44. Linoleic acid (18:2, Δ 9,12) is classified as an:

- A. ω -3 fatty acid
- B. ω -6 fatty acid
- C. ω -9 fatty acid
- D. ω -12 fatty acid

Answer: B

45. Which notation correctly describes palmitoleic acid (16:1, Δ 9) in the ω -system?

- A. ω -9
- B. ω -7
- C. ω -3
- D. ω -1

Answer: B

46. Why are certain fatty acids considered "essential" for humans?

- A. We use them for energy
- B. We lack the enzymes to synthesize them (specifically past C9)
- C. They are the only fats that can be stored
- D. They are required for protein synthesis

Answer: B

47. Which of the following is a strictly essential fatty acid that must be obtained from plants?

- A. Palmitic acid
- B. Oleic acid
- C. Linoleic acid
- D. Stearic acid

Answer: C

48. α -Linolenic acid is an essential fatty acid that serves as a precursor for:

- A. Prostaglandins
- B. ω -3 fatty acids (like EPA and DHA)
- C. Glucose
- D. Saturated fats

Answer: B

49. Arachidonic acid becomes "essential" only if there is a deficiency of:

- A. α -Linolenic acid
- B. Linoleic acid
- C. Vitamin C
- D. Glucose

Answer: B

50. A clinical symptom of essential fatty acid deficiency is:

- A. Hyperglycemia
- B. Scaly dermatitis
- C. Scurvy
- D. Excessive blood clotting

Answer: B

51. Besides dermatitis, essential fatty acid deficiency can lead to:

- A. High blood pressure
- B. Visual and neurologic abnormalities
- C. Brittle bones
- D. Increased muscle mass

Answer: B

52. Non-essential fatty acids are those that:

- A. Are not needed by the body
- B. Can be synthesized by the body from other precursors
- C. Are only found in fish
- D. Cannot be oxidized

Answer: B

53. The body can synthesize non-essential fatty acids from excess dietary:

- A. Carbohydrates and proteins
- B. Fiber
- C. Water
- D. Cholesterol

Answer: A

54. Essential fatty acids are required for the synthesis of:

- A. Glycogen
- B. Eicosanoids
- C. Essential amino acids
- D. DNA

Answer: B

55. Which of the following is a major source of essential fatty acids in the human diet?

- A. Beef tallow
- B. Butter
- C. Vegetable oils
- D. Refined sugar

Answer: C

56. Dietary ω -3 polyunsaturated fats are known to:

- A. Increase blood pressure
- B. Suppress cardiac arrhythmias
- C. Raise LDL cholesterol
- D. Cause obesity

Answer: B

57. Which ω -3 fatty acids are found predominantly in fish oils?

- A. Linoleic and arachidonic acid
- B. Eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA)
- C. Palmitic and stearic acid
- D. Oleic and palmitoleic acid

Answer: B

58. Intake of ω -3 PUFAs is associated with a reduction in:

- A. Serum triacylglycerols
- B. HDL levels
- C. Blood glucose
- D. Protein absorption

Answer: A

59. PUFAs from the ω -6 family, principally linoleic acid, have what effect when substituted for saturated fats?

- A. They raise LDL
- B. They lower plasma cholesterol and LDL
- C. They have no effect on CHD risk
- D. They prevent the absorption of Vitamin A

Answer: B

60. A potential drawback of very high ω -6 intake (above 10% of calories) is:

- A. Weight gain
- B. Concern that oxidation may lead to deleterious products
- C. Developing scurvy
- D. Reduced insulin secretion

Answer: B

61. ω -3 long-chain PUFAs are added to infant formulas to promote:

- A. Strong bones
- B. Brain development
- C. Faster digestion of lactose
- D. Early walking

Answer: B

62. The "Mediterranean diet" is considered heart-healthy because it is high in:

- A. Saturated fats from red meat
- B. Monounsaturated fatty acids (from olive oil) and ω -3s
- C. Refined carbohydrates
- D. Trans-fatty acids

Answer: B

63. ω -3 fatty acids decrease the tendency for "thrombosis," which means they:

- A. Lower blood sugar
- B. Reduce blood clot formation
- C. Increase appetite
- D. Strengthen the heart muscle

Answer: B

64. Acceptable Macronutrient Distribution Range (AMDR) for α -linolenic acid is:

- A. 45–65% of total calories
- B. 20–35% of total calories
- C. 0.6–1.2% of total calories
- D. 10–15% of total calories

Answer: C

65. Which fatty acid is the direct precursor of the eicosanoid family (prostaglandins, etc.)?

- A. Palmitic acid
- B. Oleic acid
- C. Arachidonic acid
- D. Stearic acid

Answer: C

66. Triacylglycerols are also known as:

- A. Phospholipids

- B. Triglycerides
- C. Steroids
- D. Glycolipids

Answer: B

67. In a "mixed" triacylglycerol, the fatty acid on Carbon 2 is typically:

- A. Saturated
- B. Unsaturated
- C. A short-chain fatty acid
- D. Attached by a peptide bond

Answer: B

68. Why are triacylglycerols a more efficient energy store than glycogen?

- A. They are more oxidized
- B. They are stored anhydrously (without water) and are more reduced
- C. They can be converted to protein
- D. They are soluble in blood

Answer: B

69. Where is the primary site of triacylglycerol storage in the body?

- A. Liver
- B. Skeletal muscle
- C. White adipose tissue
- D. Brain

Answer: C

70. The enzyme that initiates the mobilization of fatty acids from stored TAGs in adipose tissue is:

- A. Pancreatic lipase
- B. Lipoprotein lipase
- C. Hormone-sensitive lipase
- D. Gastric lipase

Answer: C

71. What activates hormone-sensitive lipase during fasting or exercise?

- A. Insulin

- B. Epinephrine or glucagon
- C. High blood glucose
- D. Presence of chylomicrons

Answer: B

72. What is the fate of the "glycerol" released from triacylglycerol degradation?

- A. It is converted to fatty acids in the adipose
- B. It is used by the liver for gluconeogenesis
- C. It is excreted in the urine
- D. It is used by the brain for energy

Answer: B

73. In the small intestine, TAGs are hydrolyzed by pancreatic lipase into:

- A. Free glycerol and three fatty acids
- B. 2-monoacylglycerol and two free fatty acids
- C. Cholesterol and phospholipids
- D. Amino acids

Answer: B

74. Triacylglycerols synthesized in the liver are packaged into which lipoprotein for transport?

- A. Chylomicrons
- B. LDL
- C. VLDL
- D. HDL

Answer: C

75. A patient with "creamy" plasma and TAG levels over 2,000 mg/dl likely has a deficiency in:

- A. HMG-CoA reductase
- B. Lipoprotein lipase or Apo C-II
- C. Insulin
- D. Glucagon

Answer: B

76. Phospholipids are the predominant lipids of:

- A. Adipose droplets
- B. Cell membranes
- C. Bile stones
- D. The nucleus only

Answer: B

77. The simplest glycerophospholipid, which is a precursor for others, is:

- A. Phosphatidylcholine
- B. Phosphatidylethanolamine
- C. Phosphatidic acid
- D. Sphingomyelin

Answer: C

78. Which phospholipid is the major component of lung surfactant?

- A. Phosphatidylinositol
- B. Dipalmitoylphosphatidylcholine
- C. Cardiolipin
- D. Sphingomyelin

Answer: B

79. Insufficient production of lung surfactant causes which clinical condition?

- A. Cystic fibrosis
- B. Respiratory distress syndrome (RDS)
- C. Atherosclerosis
- D. Tay-Sachs disease

Answer: B

80. Which phospholipid is virtually exclusive to the inner mitochondrial membrane?

- A. Phosphatidylserine
- B. Cardiolipin
- C. Plasmalogen
- D. Ganglioside

Answer: B

81. Phosphatidylinositol (PI) often contains which fatty acid at Carbon 2?

- A. Palmitic acid

- B. Arachidonic acid
- C. Stearic acid
- D. Oleic acid

Answer: B

82. The degradation of membrane PIP₂ by Phospholipase C produces which second messengers?

- A. cAMP and cGMP
- B. IP₃ and diacylglycerol (DAG)
- C. Glucose and Oxygen
- D. Cholesterol and Bile salts

Answer: B

83. Phospholipids that contain an ether linkage instead of an ester linkage are called:

- A. Ceramides
- B. Plasmalogens
- C. Cerebrosides
- D. Sulfatides

Answer: B

84. Which enzyme releases arachidonic acid from membrane phospholipids for eicosanoid synthesis?

- A. Phospholipase A₁
- B. Phospholipase A₂
- C. Phospholipase C
- D. Sphingomyelinase

Answer: B

85. A deficiency in the synthesis of GPI anchors in hematopoietic cells results in:

- A. Niemann-Pick disease
- B. Paroxysmal nocturnal hemoglobinuria
- C. Type 1 diabetes
- D. Gaucher disease

Answer: B

86. Sphingolipids are built on a backbone of:

- A. Glycerol
- B. Sphingosine
- C. Cholesterol
- D. Inositol

Answer: B

87. Sphingosine is derived from the amino acid serine and which fatty acid?

- A. Stearate
- B. Palmitate
- C. Oleate
- D. Linoleate

Answer: B

88. A "ceramide" is formed by attaching a fatty acid to the amino group of sphingosine via an:

- A. Ester linkage
- B. Amide linkage
- C. Glycosidic bond
- D. Phosphodiester bridge

Answer: B

89. Sphingomyelin is the only significant human sphingophospholipid and contains:

- A. A sugar residue
- B. Phosphorylcholine attached to ceramide
- C. A sulfate group
- D. Three fatty acids

Answer: B

90. Cerebrosides are the simplest neutral glycolipids and contain:

- A. An oligosaccharide
- B. A single sugar molecule (e.g., galactose or glucose)
- C. Sialic acid
- D. Phosphorus

Answer: B

91. Gangliosides are the most complex glycosphingolipids and are characterized by the

presence of:

- A. Choline
- B. N-acetylneuraminic acid (NANA or sialic acid)
- C. Sulfate groups
- D. Glycerol

Answer: B

92. Where are glycosphingolipids found in the highest concentrations?

- A. Adipose tissue
- B. Myelin sheath and nerve tissue
- C. Liver
- D. Heart muscle

Answer: B

93. Hereditary deficiencies of lysosomal hydrolases that degrade sphingolipids cause:

- A. Hyperlipidemias
- B. Sphingolipidoses (lipid storage diseases)
- C. Diabetes
- D. Scurvy

Answer: B

94. Which disease is caused by a deficiency in sphingomyelinase, leading to sphingomyelin accumulation?

- A. Tay-Sachs
- B. Gaucher
- C. Niemann-Pick (A + B)
- D. Fabry

Answer: C

95. Gaucher disease, the most common sphingolipidosis, involves the accumulation of:

- A. Gangliosides
- B. Glucocerebrosides
- C. Sphingomyelin
- D. Cholesterol

Answer: B

96. Cholesterol is the parent compound for all of the following EXCEPT:

- A. Bile acids
- B. Steroid hormones
- C. Vitamin D
- D. Phospholipids

Answer: D

97. Steroids with a hydroxyl group at Carbon 3 and a side chain at Carbon 17 are called:

- A. Ketones
- B. Sterols
- C. Esters
- D. Eicosanoids

Answer: B

98. Which derived lipid class mediates the inflammatory response and is inhibited by NSAIDs like aspirin?

- A. Ketone bodies
- B. Prostaglandins
- C. Bile salts
- D. Cholesterol

Answer: B

99. The rate-limiting step in cholesterol synthesis is catalyzed by:

- A. HMG-CoA reductase
- B. Acetyl CoA carboxylase
- C. Fatty acid synthase
- D. Lipoprotein lipase

Answer: A

100. A high content of saturated fat and trans fat in the diet tends to increase levels of:

- A. HDL cholesterol
- B. LDL cholesterol
- C. Blood glucose
- D. Dietary fiber

Answer: B

